A spatial analysis of SST & Chlorophyll Concentration of North Sea

Abstract

Sea Surface Temperature (SST) refers to the temperature of the water near the ocean surface, which mainly varies with latitude; the warmest regions are generally located at low latitudes. Chlorophyll concentration indicates the amount of photosynthetic plankton or phytoplankton present in the ocean. Both sea surface temperature and chlorophyll concentration are important parameters for understanding ocean characteristics.

SST and chlorophyll concentration may vary on daily, monthly, and seasonal scales. This study aims to examine the spatial and temporal variations of sea surface temperature and chlorophyll concentration in the North Sea using Geographic Information System (GIS) techniques. Satellite data from multiple organizations were used and analyzed with ArcGIS software.

The results show that the highest sea surface temperature was observed in the German region, while the lowest was found in the UK region. The highest average chlorophyll concentrations were recorded in the English Channel, and the German and Danish parts, whereas the lowest concentrations occurred in the UK and Norwegian parts. The distribution of sea surface temperature and chlorophyll, derived through remote sensing, can help identify potential fishing zones. This study also attempts to determine the potential fishing grounds for a fish species called *sand eel*.

Keywords

Sea Surface Temperature (SST); Chlorophyll Concentration; North Sea; Remote Sensing; GIS; ArcGIS; Satellite Data; Potential Fishing Zone; Sand Eel

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1. Introduction

The North Sea is a semi-enclosed, shallow shelf sea located in northwestern Europe and is considered one of the most productive seas in the world. It lies between Great Britain, Norway, Denmark, Germany, the Netherlands, Belgium, and France, forming part of the Atlantic Ocean. The North Sea connects to the ocean through the English Channel and the Norwegian Sea, making it an epeiric or "shelf" sea on the European continental shelf.

Covering an area of about 570,000 square kilometers, it extends approximately 970 kilometers (600 miles) in length and 580 kilometers (360 miles) in width. The surrounding countries are highly industrialized and agriculturally active, contributing significant amounts of nutrients and pollutants to the North Sea. Despite this, it has traditionally supported a large fishing industry and has played an important role in the maritime history of Northern Europe.

The North Sea shows significant variation in sea surface temperature (SST) and chlorophyll concentration, which may fluctuate monthly, seasonally, and annually. In this study, we aim to analyze the monthly variation of sea surface temperature and chlorophyll concentration in the North Sea. Additionally, the study seeks to identify the potential fishing grounds for *sand eel* using remote sensing and GIS techniques.

1.1. Objectives

1) To find out the monthly SST (Sea Surface Temperature) variation of North Sea.
2) To find out the Chlorophyll concentration variation of North Sea.
3) To find out the potential fishing ground of Sand ell.

2. METHODS

2.1. Study area:

The North Sea is a marginal sea of the Atlantic Ocean, located between Great Britain, Norway, Denmark, Germany, the Netherlands, Belgium, and France. It is an epeiric (or "shelf") sea situated on the European continental shelf, connecting to the Atlantic Ocean through the English Channel in the south and the Norwegian Sea in the north. The North Sea extends over 970 kilometers (600 miles) in length and 580 kilometers (360 miles) in width, covering an area of approximately 570,000 square kilometers (220,000 square miles).
Coordinate of our study area (North sea) is

Min. Lat: 50° 59' 43.3" N (50.9954°)
Min. Long: 4° 26' 43.3" W (-4.4454°)
Max. Lat: 61° 1' 1.3" N (61.017°)
Max. Long: 12° 0' 21.4" E (12.0059°)

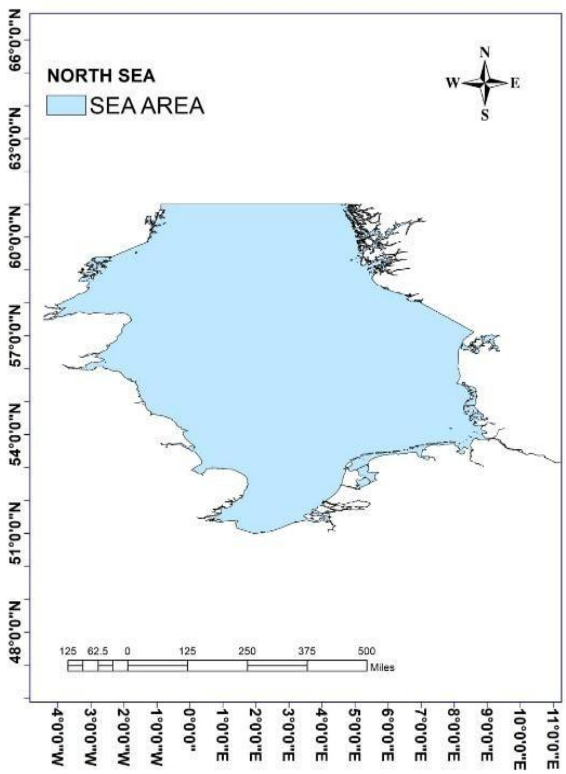

Figure 1: North Sea

2.2. Data collection:

We collect our data from some web site. They are Marineregions.org, ERDDAP, National Centers for Environmental Information & Ocean color web. The links are given below:

1) https://www.marineregions.org
2) https://coastwatch.pfeg.noaa.gov/erddap/index.html
3) https://www.ncei.noaa.gov/products
4) https://oceancolor.gsfc.nasa.gov

3. Project activities

3.1. SST

Sea Surface Temperature (SST), also known as ocean surface temperature, refers to the temperature of the water near the ocean's surface. SST varies across different seasons; during warmer months, the temperature is generally higher than in winter. In this study, we used satellite data for two months—April 2021 and May 2021—for the North Sea. The mean value of sea surface temperature was then calculated based on these two months of data.

Data interpolation:

The method of data interpolation is given below:

- ➢ Open Arc GIS
- ➢ Make IHO
- ➢ Input data
- ➢ Spatial Analyst too
- ➢ Interpolation
- ➢ Kriging
- ➢ Input point feature
- ➢ Select environment
- ➢ Processing extent, raster analysis
- ➢ Done
- ➢ Export map

Our graphs are given below.

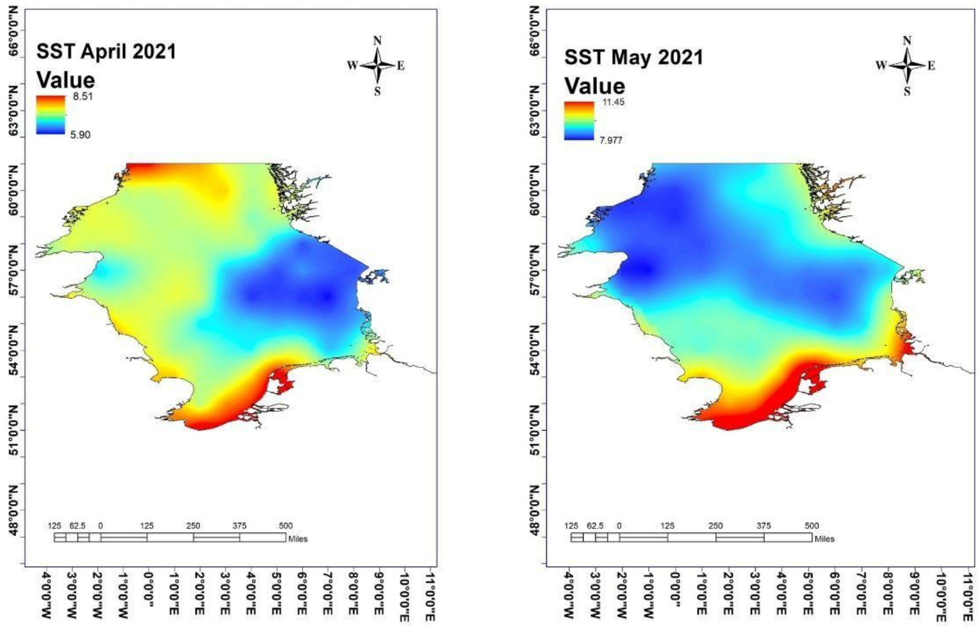

Figure 2: SST of April & May month 2021

Mean of this 2 months data is given below:

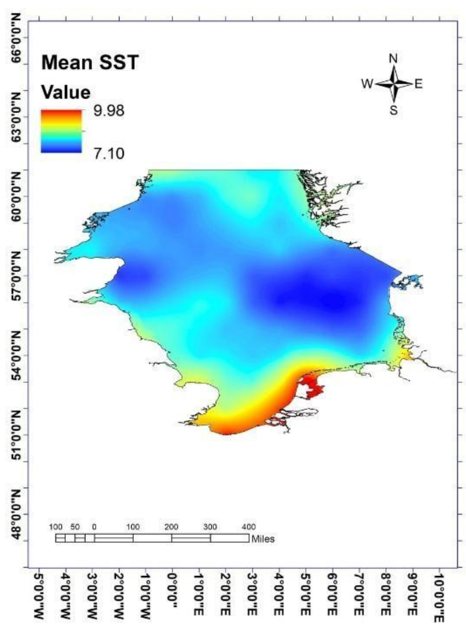

Figure 3: Mean SST of April & May 2021

We can classified mean SST to find which regions are warmer than other study point. Classification of mean SST of April & May are given below.

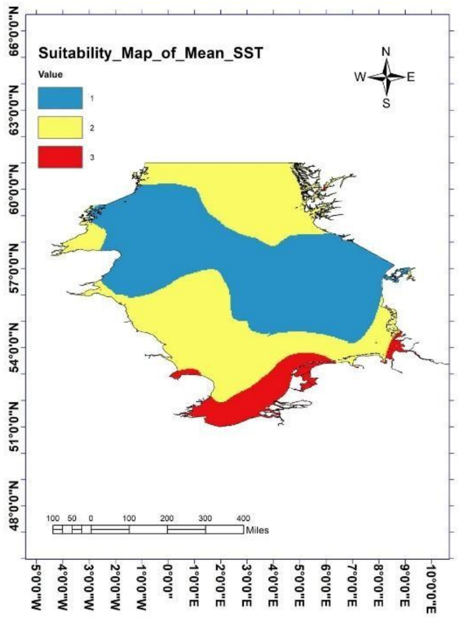

Figure 4: Reclassification of mean SST

3.2. Chlorophyll Concentration:

The concentration of chlorophyll is an indicator of the amount of photosynthetic plankton, or phytoplankton, present in the ocean. Phytoplankton populations are influenced by various climatic factors, such as sea surface temperature and wind patterns. Chlorophyll concentration is widely used to measure the abundance of these microscopic organisms, as the ocean is largely composed of such biological components.
In this study, we used satellite data for two months—April 2021 and May 2021—for the North Sea. The mean chlorophyll concentration was then calculated based on these two months of data.

Data interpolation:

The method of data interpolation is given below:

- ➢ Open Arc GIS
- ➢ Make IHO
- ➢ Input data
- ➢ Spatial Analyst tool
- ➢ Interpolation
- ➢ Kriging

➢ Input point feature
➢ Select environment
➢ Processing extent, raster analysis
➢ Done
➢ Export map

Our graphs are given below:

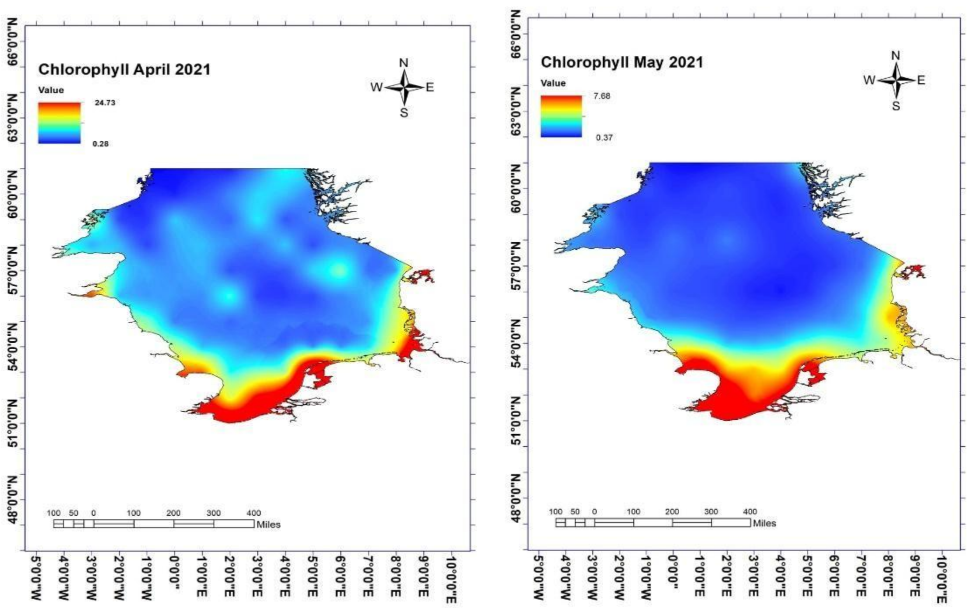

Figure 5 : Chlorophyll of April & May month 2021

Mean of this 2months data is given below:

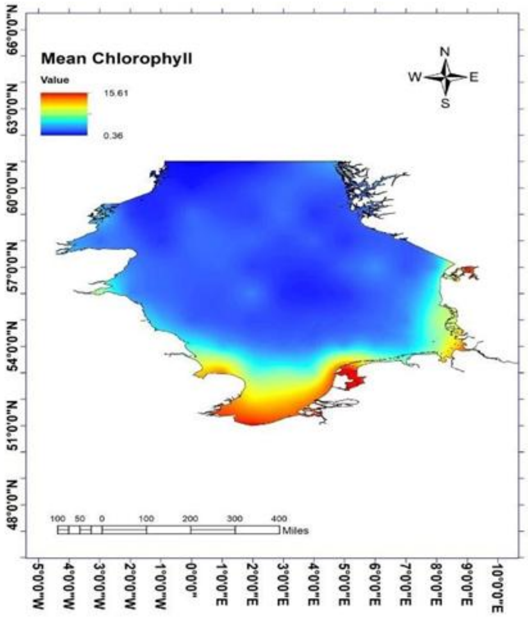

Figure 6: Mean SST of April & May 2021

Classification of mean SST of April & May are given below.

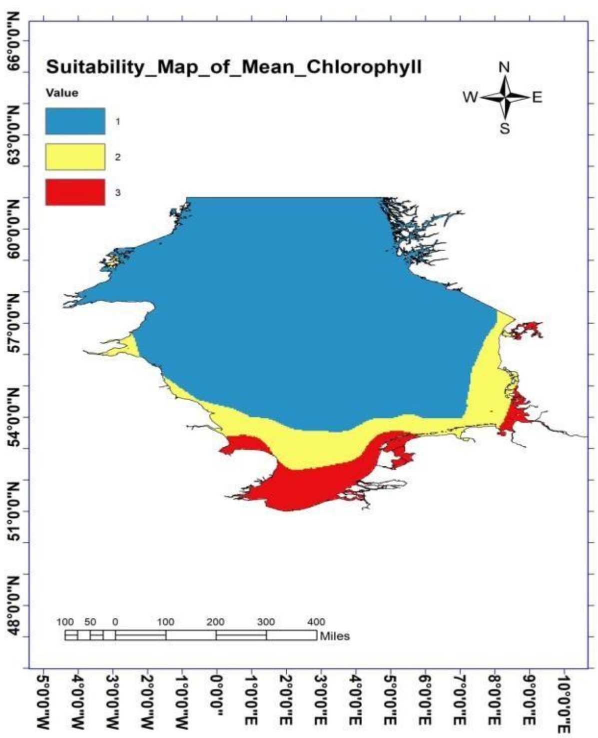

Figure 7 : Reclassification of mean SST

Determination of potential fishing ground: with using georeferencing we find a map of potential fishing ground of sand ell. The map is given below:

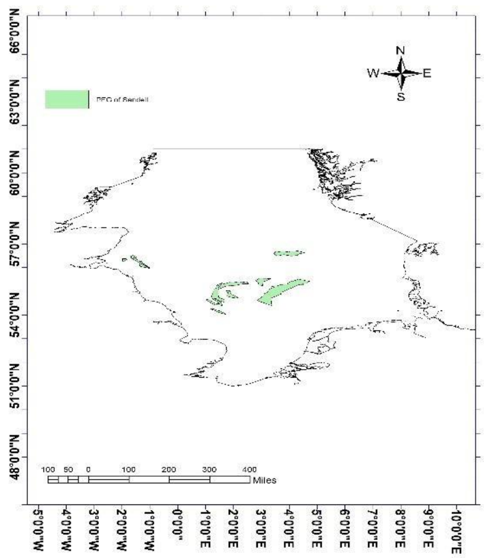

Figure 8: potential fishing ground of sand ell

4. Results

4.1. Sea Surface Temperature

In our study area (the North Sea), we aim to identify the variation in Sea Surface Temperature (SST). SST can vary monthly, seasonally, annually, and even over decades. To analyze the SST variation of the North Sea, we collected satellite data and created interpolated maps.
From the SST graph of April 2021, the maximum temperature was observed in the UK (North Atlantic region) and the German parts of the North Sea, ranging from 8°C to 8.5°C. The Danish region appeared colder, with temperatures around 6°C, while the mid-ocean region recorded approximately 7°C.
In May 2021, the warmest region was again the German part of the North Sea (around 11°C), whereas the UK region remained cooler at about 8°C. The mid-region recorded around 10°C.
In the mean SST map, the German part was the warmest (around 10°C), while the Danish region was relatively cooler (around 7°C). If we classify the sea based on SST variation, it can be divided into warm, moderate, and cold zones.

- ➢ UK & Danish part is Cold region.
- ➢ Norway & English Channel part is warmer than UK & Danish part.
- ➢ German part is most warmer than any other part.

4.2. Chlorophyll

In our study area, we also analyzed the chlorophyll concentration of the North Sea. Chlorophyll concentration largely depends on factors such as nutrient availability, river discharge, rainfall intensity, and salinity. In the North Sea, the English Channel and Norwegian Sea serve as major sources of river discharge. As a result, these regions exhibit relatively higher chlorophyll concentrations.

From the chlorophyll concentration map of April 2021, it was observed that the English Channel, the German region, and the Norwegian Sea had higher chlorophyll concentrations, ranging around 24 mg/L. In contrast, the UK part and the North Atlantic Oceanic region showed relatively lower concentrations, ranging between 0.25 and 1 mg/L.

In May 2021, the chlorophyll concentration varied noticeably. During this month, the maximum concentration reached 7.68 mg/L, while the minimum was 0.37 mg/L. High chlorophyll abundance was found in the German region and the English Channel (around 8 mg/L), as well as in the Norwegian region. However, the oceanic part of the North Sea showed relatively low chlorophyll values.

In the mean chlorophyll concentration map, the German region appeared to be the richest in chlorophyll abundance, whereas the North Atlantic part remained the poorest.

If we classify the chlorophyll concentration of the North Sea into three categories, we find:

1. High concentration zone – German, Danish regions, and the English Channel
2. Moderate concentration zone – Norwegian region
3. Low concentration zone – UK region and North Atlantic Oceanic part

Potential Fishing Ground of Sand El

In this project, we also attempted to identify the potential fishing grounds (PFG) for *sand eel* fish species. Using georeferenced data, we mapped the suitable habitats for *sand eel* in the North Sea. The results indicated that the potential fishing grounds are mainly located near the German coast and the English Channel region. A total of six potential fishing grounds for *sand eel* were identified in this study.

5. Discussion

From our project findings, it can be observed that the German part of the North Sea is comparatively warmer than the other regions and shows a high abundance of chlorophyll concentration. In contrast, the UK and the North Atlantic Ocean regions exhibit relatively lower temperatures and low chlorophyll concentrations. The central region of the North Sea represents a moderate condition, with medium levels of both sea surface temperature (SST) and chlorophyll concentration.

The results indicate a strong relationship between temperature and chlorophyll distribution, suggesting that warmer areas tend to support higher phytoplankton growth. Furthermore, the potential fishing grounds for *sand eel* were identified mainly near the German coast and the English Channel region, where both SST and chlorophyll concentration are relatively high. These findings highlight the importance of SST and chlorophyll data in identifying productive marine zones and understanding the ecological dynamics of the North Sea.

6. Conclusion

In this study, we analyzed the monthly variations of sea surface temperature (SST) and chlorophyll concentration in the North Sea. Significant spatial and temporal variations were observed in both SST and chlorophyll levels across different regions and months. The German region and the English Channel were found to be comparatively warmer and exhibited higher chlorophyll concentrations than other parts of the North Sea. These findings suggest that regions with higher temperatures tend to support greater phytoplankton abundance. Overall, this study highlights the importance of SST and chlorophyll data in understanding the ecological dynamics of the North Sea and identifying potential productive zones, such as the potential fishing grounds for *sand eel* near Germany and the English Channel.